Project 1: Database Normalization Design

Functional Dependencies & BCNF Relations

-Note: I organized the sections but all the information is there.

Section 1: Functional Dependencies

The following represents the canonical cover for the dataset. Each left-hand side is irreducible, and no dependency is derivable from the others.

1A: Metadata Dependencies

- {agency_code} -> {agency_name, agency_abbr}
- {loan_type} ->{loan_type_name}
- {property_type} ->{property_type_name}
- {loan_purpose} ->{loan_purpose_name}
- {owner_occupancy} ->{owner_occupancy_name}
- {preapproval} ->{preapproval_name}
- {action_taken} ->{action_taken_name}
- {msamd} ->{msamd_name}
- {state_code} ->{state_name, state_abbr}
- {county_code} ->{county_name}
- {applicant_ethnicity} ->{applicant_ethnicity_name}
- {co_applicant_ethnicity} ->{co_applicant_ethnicity_name}
- {applicant_sex} ->{applicant_sex_name}
- {co_applicant_sex} ->{co_applicant_sex_name}
- {applicant_race_1} ->{applicant_race_name_1}
- {purchaser_type} ->{purchaser_type_name}
- {denial_reason_1} ->{denial_reason_name_1}
- {hoepa_status} ->{hoepa_status_name}
- {lien_status} ->{lien_status_name}
- {edit_status} ->{edit_status_name}

1B: Location Composite Dependency

- {county_code, msamd, state_code, census_tract_number, population, minority_population, hud_median_family_income, tract_to_msamd_income, number_of_owner_occupied_units, number_of_1_to_4_family_units} ->{Location_ID}

1C: Row Identity

- {ID} ->{as_of_year, respondent_id, agency_code, loan_type, property_type, loan_purpose, owner_occupancy, loan_amount_000s, preapproval, action_taken, Location_ID, applicant_ethnicity, co_applicant_ethnicity, applicant_sex, co_applicant_sex, applicant_income_000s, purchaser_type, rate_spread, hoepa_status, lien_status, edit_status, sequence_number, application_date_indicator}

Section 2: Normalized Relations (BCNF)

These relations are in Boyce-Codd Normal Form. The decomposition is lossless and dependency-preserving, ensuring full recovery of the original dataset.

2A: Main Application Table

- Application(ID, as_of_year, respondent_id, agency_code, loan_type, property_type, loan_purpose, owner_occupancy, loan_amount_000s, preapproval, action_taken, Location_ID, applicant_ethnicity, co_applicant_ethnicity, applicant_sex, co_applicant_sex, applicant_income_000s, purchaser_type, rate_spread, hoepa_status, lien_status, edit_status, sequence_number, application_date_indicator)

2B: Location Table

- Location(Location_ID, county_code, msamd, state_code, census_tract_number, population, minority_population, hud_median_family_income, tract_to_msamd_income, number_of_owner_occupied_units, number_of_1_to_4_family_units)

2C: 1st Normal Form Join Tables

- Applicant_Race(ID, race_code, race_number)
- Co_Applicant_Race(ID, race_code, race_number)
- Denial_Reasons(ID, denial_reason_code, reason_number)

2D: Lookup Tables

- Agency(agency_code, agency_name, agency_abbr)
- LoanType(loan_type, loan_type_name)
- PropertyType(property_type, property_type_name)
- LoanPurpose(loan_purpose, loan_purpose_name)
- OwnerOccupancy(owner_occupancy, owner_occupancy_name)
- Preapproval(preapproval, preapproval_name)
- ActionTaken(action_taken, action_taken_name)

- MSAMD(msamd, msamd_name)
- State(state_code, state_name, state_abbr)
- County(county_code, county_name)
- Ethnicity(ethnicity_code, ethnicity_name)
- Sex(sex_code, sex_name)
- Race(race_code, race_name)
- PurchaserType(purchaser_type, purchaser_type_name)
- DenialReason(denial_reason_code, denial_reason_name)
- HoepaStatus(hoepa_status, hoepa_status_name)
- LienStatus(lien_status, lien_status_name)
- EditStatus(edit_status, edit_status_name)
    - But this is actually just null so these are like placeholders

Notes:

- Primary Keys: ID is the unique primary key for applications. Location_ID is the primary key for all location-related data.
- Integrity: Only respondent_id remains as text in the main table; all other fields are numeric or integer types.
- Lossless Decomposition: All original attributes are retained to allow for perfect regeneration of the original source CSV.